Step-Up Fabrication Document

Top Fabrication (Scale 1:1)

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

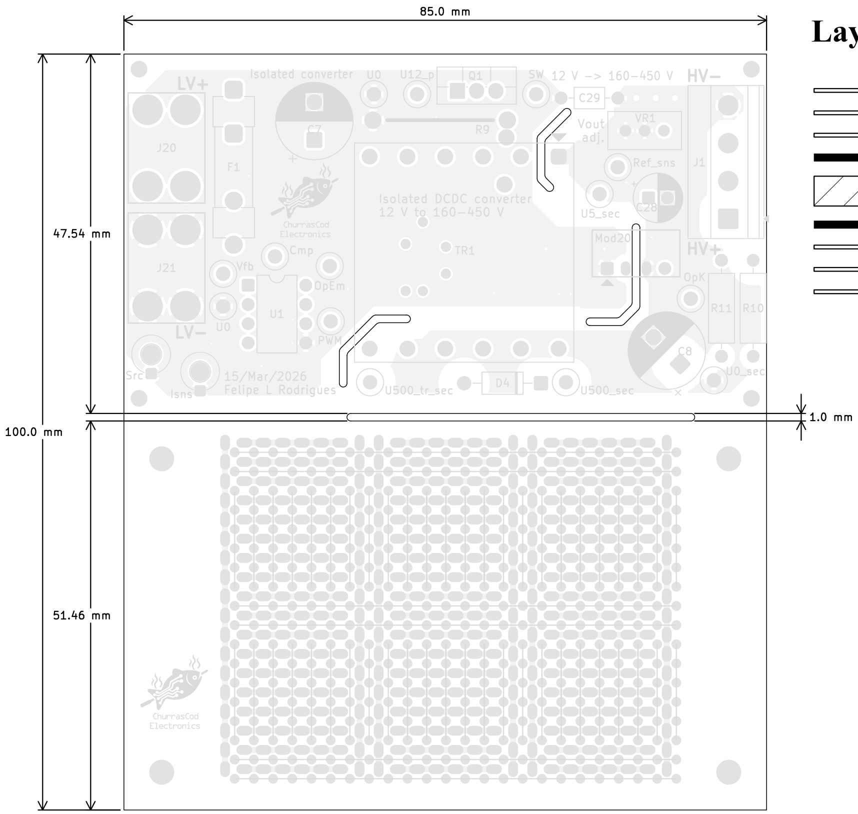
1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

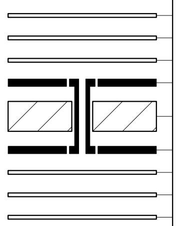
   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
4) SURFACE FINISH: HAL LEAD-FREE
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR YELLOW.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY BUYER.
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	85.000 × 100.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.300 mm
TRACE TO TRACE	0.020 mm
MIN. HOLE (PTH)	0.250 mm
MIN. HOLE (NPTH)	2.100 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.200 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm



Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02 mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035 mm (1 oz)**		**Signal**	**GBR**
Core		1.49 mm	FR4	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.035 mm (1 oz)**		**Signal**	**GBR**
	B.Mask	0.02 mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6 mm

Note: external layer thicknesses are specified after plating

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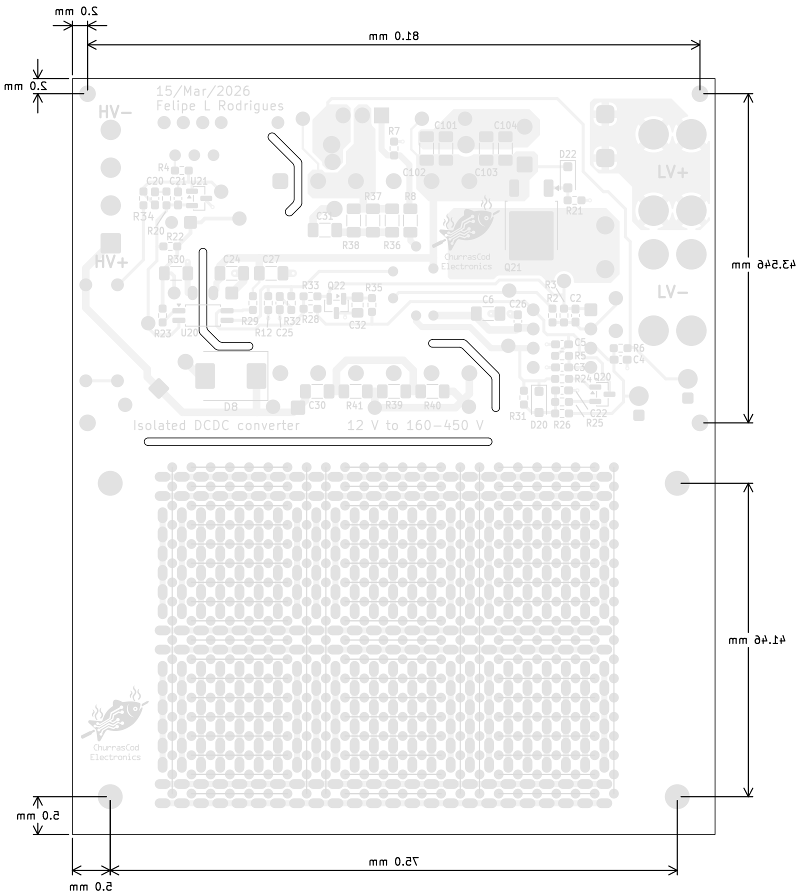
Comments:	Company: NA		Variant: CHECKED	Git Hash: 26c8ee0
	Board Name: **Step-Up**		Project Name: **Isolated step-up 12:450 V**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: **A4**	Sheet: **1** of **7**

Step-Up Fabrication Document

Bottom Fabrication (Scale 1:1)



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Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2026-02-07	Revision: 2.0.0+ (Unreleased)
Sheet Path:		Reviewer: FR	Size: **A4**	Sheet: **2** of **7**

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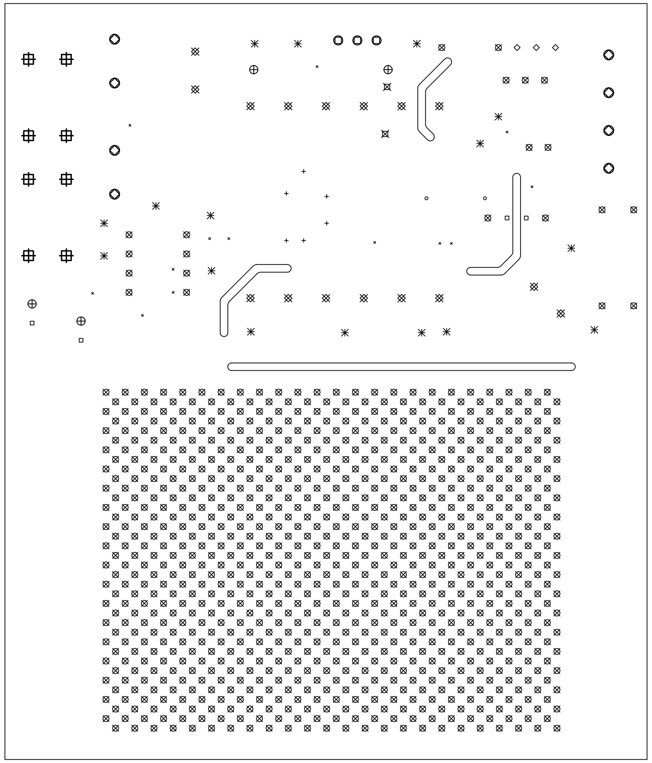
Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	13	0.25mm (9.84mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
○	2	0.40mm (15.75mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
+	6	0.50mm (19.69mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
□	4	0.70mm (27.56mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
◇	3	0.80mm (31.50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
⦻	885	0.80mm (31.50mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
✳	16	0.90mm (35.43mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⌧	2	1.00mm (39.37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Via
※	16	1.00mm (39.37mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊕	4	1.10mm (43.31mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊡	3	1.20mm (47.24mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⟐	8	1.30mm (51.18mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
⊞	8	1.80mm (70.87mils)	PTH	Round	L1 (Ref) - L2 (Sig, PWR)	Pad
	Total 970					



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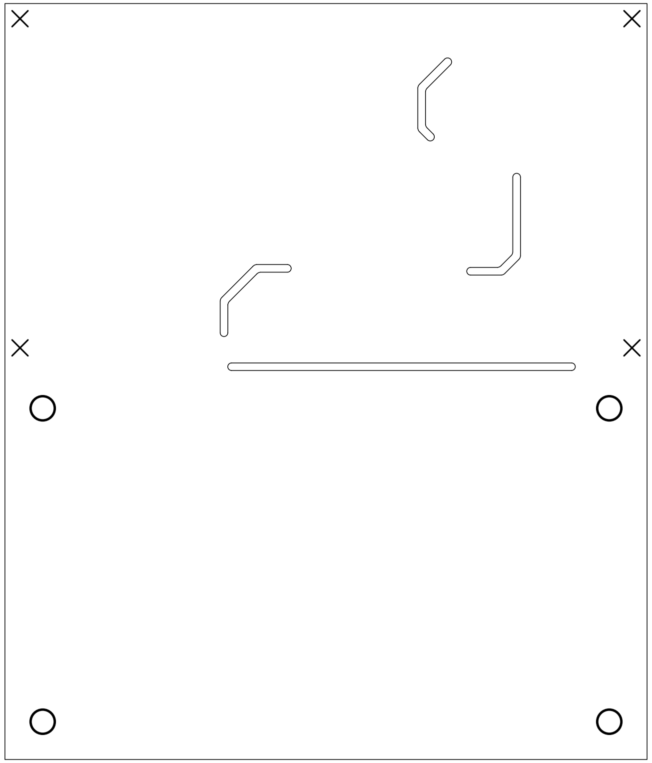
Drill Drawing L1 - L4 (Scale 1:1)

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
○	4	3.20mm (125.98mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 8					



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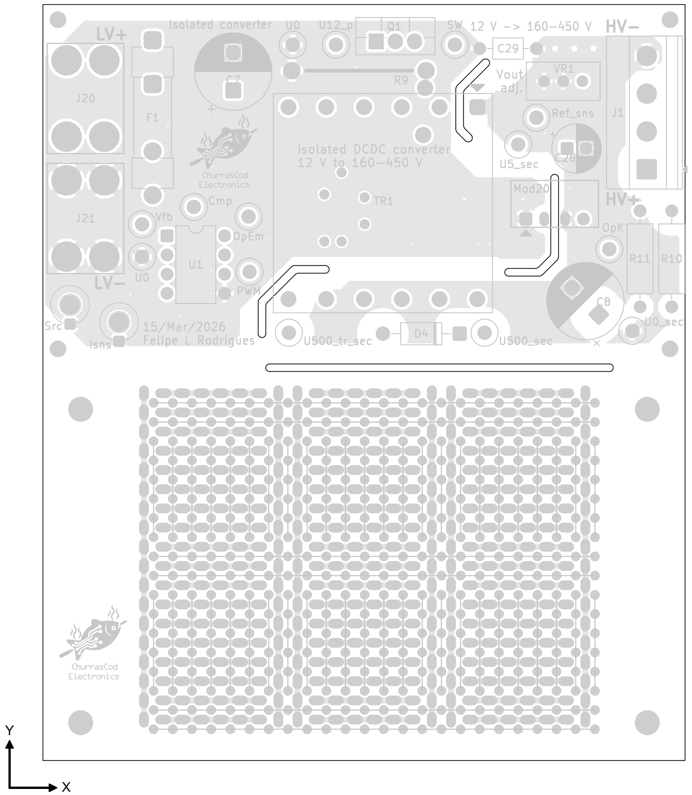
Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP9	Sec_ref_sns	69.73	88.65
TP10	Sec_OpK	79.38	71.25
TP20	SW	58.93	98.30
TP23	PWM	31.76	68.26
TP24	Vfb	17.53	74.55
TP25	Comp	24.39	76.83
TP26	OpEm	31.63	75.56
TP27	U500_sec	62.87	60.20
TP28	U0_sec	82.43	60.45
TP30	U500_tr_sec	36.96	60.20
TP31	U12_p	43.19	98.30
TP32	U0	17.53	70.23
TP33	U0	37.47	98.30
TP34	U5_sec	67.32	85.09
TP35	Isns	14.48	60.34
TP36	Src,U0	8.01	62.61
TP101	Net-(TP101-Pad1)	20.33	52.20
TP102	Net-(TP101-Pad1)	22.87	52.20
TP103	Net-(TP101-Pad1)	25.41	52.20
TP104	Net-(TP101-Pad1)	27.95	52.20
TP105	Net-(TP101-Pad1)	30.49	52.20
TP106	Net-(TP101-Pad1)	33.03	52.20
TP107	Net-(TP107-Pad1)	20.33	49.66
TP108	Net-(TP107-Pad1)	22.87	49.66
TP109	Net-(TP107-Pad1)	25.41	49.66
TP110	Net-(TP107-Pad1)	27.95	49.66
TP111	Net-(TP107-Pad1)	30.49	49.66
TP112	Net-(TP107-Pad1)	33.03	49.66
TP113	Net-(TP113-Pad1)	20.33	47.12
TP114	Net-(TP113-Pad1)	22.87	47.12
TP115	Net-(TP113-Pad1)	25.41	47.12
TP116	Net-(TP113-Pad1)	27.95	47.12

All dimensions are in millimeters
unless otherwise specified.



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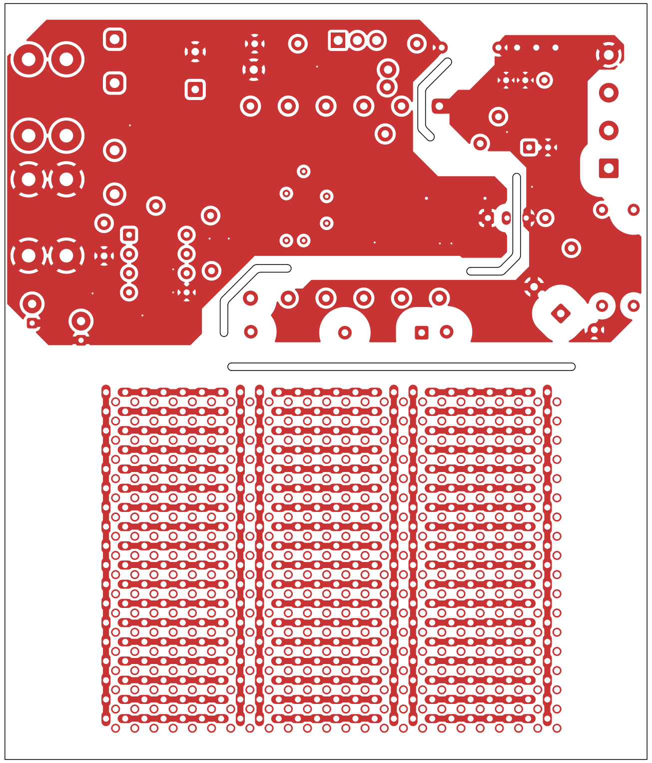
Step-Up Fabrication Document

F.Cu (Scale 1:1)



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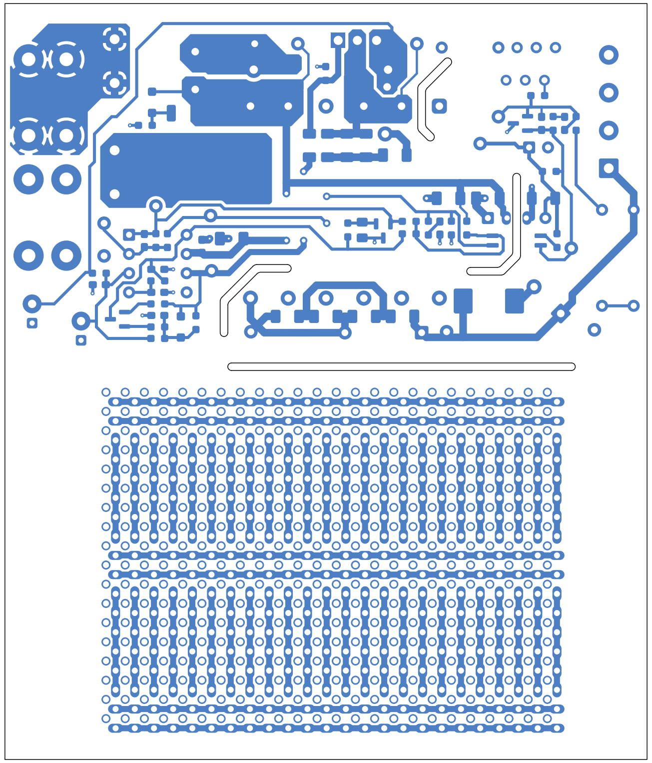
B.Cu (Scale 1:1)



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Sheet Path:		Reviewer: FR	Size: A4	Sheet: 7 of 7